

Pre-Board Exam (2024-25)
Class: X
Subject: Artificial Intelligence (417)

Time: 2 hours

Max Marks: 50

MARKING SCHEME

SECTION A: OBJECTIVE TYPE QUESTIONS

Q. No.	Answer	Marks
Q. 1		
i.	b) External Motivation	1
ii.	a) Selected	1
iii.	d) A is not correct but R is correct	1
iv.	c) Both of the above	1
v.	c) Cultural barriers	1
vi.	(d) Reduced inequalities	1
Q. 2		
i.	(a) Both A and R are correct and R is the correct explanation of A	1
ii.	(b) Data Privacy	1
iii.	(c) Both Statement1 is incorrect and Statement2 are correct	1
iv.	Larger	1
v.	(c) csv	1
vi.	(a) stop word	1
Q. 3		
i.	(b) I, ii and iv	1

ii.	(d) System hacking	1
iii.	(d) All of the above	1
iv.	(b) To understand visual information	1
v.	(b) Smart Bot	1
vi.	(b) Precision and Recall	1
Q. 4		
i.	(C) Deep Learning	1
ii.	(c) Statement1 is correct but Statement2 is incorrect	1
iii.	(b) high precision high recall	1
iv.	(c) Classification and localization	1
v.	(b) Sentiment Analysis	1
vi.	Overfitting	1
Q. 5		
i.	(b) Natural Language Processing	1
ii.	Recall	1
iii.	(a) Categorizing each pixel in the image in to a specific class	1
iv.	(c) Automatic summarization	1
v.	(c) Silvi	1
vi.	(b) confusion matrix	1

SECTION B: SUBJECTIVE TYPE QUESTIONS

Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks)

Answer each question in 20 – 30 words.

Q. No.	QUESTION	Marks
Q. 6	1. Not understanding language 2. Cultural difference 3. Communication problem etc (2 marks for any two correct points from the above)	2
Q. 7	Specific, Measurable, Achievable, Relevant, and Time-Bound (1 mark for expansion only 2 for explanation)	2
Q. 8	1) Never eat anything over a keyboard. As crumbs of food can damage the internal part of a keyboard. 2) Always clean a Keyboard with a soft brush to remove crumbs and dust particles. 3) Wipe the screen with a soft cloth to remove any finger marks. 4) Avoid eating and keeping glasses of water near s computer as liquid split over an electronic device can spoil it beyond repair. (2 marks for any two correct points from the above)	2
Q. 9	1. Entrepreneurs are high risk takers. They are neither high nor low risk – takers. 2. They always prefer situations which lead to generating profits. 3. Entrepreneurs become rich as they start a new business. 4. Entrepreneurs have limited dedication which is the key to success. (2 marks for any two valid points)	2
Q. 10	Any two real life examples. (2 marks for any two correct points)	2
	Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)	
Q. 11	1. If it is not relevant and authentic then the testing data may go wrong or produced irrelevant predictions. 2. It increases the AI Project efficiency. (2 marks for any one correct point with explanation)	2
Q. 12	1. supervised learning uses labeled input and output data 2. supervised learning models tend to be more accurate than unsupervised learning models 3. Supervised learning is a simple method for machine learning. (2 marks for any two correct advantages)	2

Q. 13	There could be many ways and sources from where we can collect structures and reliable datasets namely Surveys, Web scrapping, Sensors, Cameras, Observations, Research, Investigation, API etc. (1 mark for each source name)	2
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Q. 14	An RGB image refers to an image that stores data for red, green, and blue color channels. These images can be stored in a single file or in multiple files, with each file representing a different wavelength. (1 mark for explanation; 1 mark for example)	2
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Q. 15	Bag of Words is a NLP model which helps in extracting features from textual data. It converts text sentences into numeric vectors by returning the unique words along with its number of occurrences. “Stop words” are the most common words in a language like “the”, “a”, “on”, “is”, “all”. These words do not carry important meaning and are usually removed from texts. (1 mark for each correct explanation)	2
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Q. 16					2
	Confusion Matrix		Reality		
			Yes	No	
	Prediction	Yes	55	10	
		No	15	20	
(½ marks each for mapping the values in the correct section, ½ *4=2 marks)					

	Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)	
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Q. 17	1 mark for AI definition 1 mark for correct AI real life application 2 marks for the ways AI machine is trained – supervised, unsupervised with explanation	4
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Q. 18	The 4Ws Problem canvas helps in identifying the key elements related to the problem. The 4Ws are Who, What, Where and Why - The “Who” block helps in analysing the people getting affected directly or indirectly due to the problem. - The “What” block helps us to determine the nature of the problem. - The “Where” block helps us to look into the situation in which the problem arises, the context of it, and the locations where it is prominent. - The “Why” block suggests to us the benefits which the stakeholders would get from the solution and how it will benefit them as well as the society <table border="1"> <tr> <td>Our</td><td>[stakeholders]</td><td>Who</td></tr> <tr> <td>Have a problem that</td><td>[need]</td><td>What</td></tr> <tr> <td>When/while</td><td>[context/ location]</td><td>Where</td></tr> </table>	Our	[stakeholders]	Who	Have a problem that	[need]	What	When/while	[context/ location]	Where	4
Our	[stakeholders]	Who									
Have a problem that	[need]	What									
When/while	[context/ location]	Where									

	Problem Statement Template <table><tr><td></td><td>/ situation]</td><td></td></tr><tr><td>An ideal solution would be</td><td>[solution]</td><td>Why</td></tr></table> (½ mark each for explanation of 4w s; 2 marksfor drawing the problem statement template with correct words in it or explaining the problem statement template) or (1 mark to be allotted if only 4Ws are written without explanation)		/ situation]		An ideal solution would be	[solution]	Why																																																																	
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Q. 19	Figure 1 Supervised learning In a supervised learning model, the dataset which is fed to the machine is labelled. In other words, we can say that the dataset is known to the person who is training the machine only then he/she is able to label the data. A label is some information which can beused as a tag for data. Figure 2 Unsupervised learning An unsupervised learning model workson unlabeled dataset. This means that the data which is fed to the machine is random and thereis a possibility that the person who is training the model does not have any information regarding it. The unsupervised learning models are used to identify relationships, patterns andtrends out of the data which is fed into it. It helps the user in understanding what the data isabout and what are the major features identified by the machine in it. (1 mark each for identifying each term supervised learning and unsupervised learning; 1 mark per explanation of each term)	4																																																																						
Q. 20	Normalization of the given text: 1) Sentence Segmentation: 1. Raj and Vijay are best friends. 2. They play together with other friends. 3. Raj likes to play football but Vijay prefers to play online games. 4. Raj wants to be a footballer. 5. Vijay wants to become an online gamer. 2) Tokenization: Tokenisation <table><tr><td>Raj and Vijay are best friends.</td><td>Raj</td><td>and</td><td>Vijay</td><td>are</td><td>best</td><td>friends</td><td>.</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>They play together with other friends.</td><td>They</td><td>play</td><td>together</td><td>with</td><td>other</td><td>friends</td><td>.</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Raj likes to play football but Vjaj prefers to play online</td><td>Raj</td><td>likes</td><td>to</td><td>play</td><td>football</td><td>but</td><td>Vijay</td><td>prefers</td><td>to</td><td>play</td><td>online</td><td>games</td><td>.</td></tr><tr><td>Raj wants to be a footballer.</td><td>Raj</td><td>wants</td><td>to</td><td>be</td><td>a</td><td>footballer</td><td>.</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Vijay wants to become an online gamer.</td><td>Vijay</td><td>wants</td><td>to</td><td>become</td><td>an</td><td>online</td><td>gamer</td><td>.</td><td></td><td></td><td></td><td></td><td></td></tr></table>	Raj and Vijay are best friends.	Raj	and	Vijay	are	best	friends	.							They play together with other friends.	They	play	together	with	other	friends	.							Raj likes to play football but Vjaj prefers to play online	Raj	likes	to	play	football	but	Vijay	prefers	to	play	online	games	.	Raj wants to be a footballer.	Raj	wants	to	be	a	footballer	.							Vijay wants to become an online gamer.	Vijay	wants	to	become	an	online	gamer	.						4
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3) Removing Stop words, Special Characters and Numbers:

In this step, the tokens which are not necessary are removed from the token list. So, the words and, are, to, an, (Punctuation) will be removed. Converting text to a common case:

4) Convert the whole text into a similar case

After the stop words removal, we convert the whole text into a similar case, preferably lower case. Here we don't have words in different case so this step is not required for given text.

5) Stemming/ Lemmatization:

In this step, the remaining words are reduced to their root words. In other words, stemming is the process in which the affixes of words are removed and the words are converted to their base form. Word Affixes Stem

Likes -s Like

Prefers -s Prefer

Wants -s want

In the given text Lemmatization is not required.

Lemmatization: Lemmatization is the process in which the affixes of words are removed and the words are converted to their base form with meaning.

Lemmatization makes sure that lemma is a word with meaning.

Finally, we reduced Given Text (Corpus) into Normalized Text.

1. raj, vijay, best, friends
2. play, together, with, other, friends
3. raj, like, play, football, but, vijay, prefer, play, online, games
4. raj, want, be, footballer
5. vijay, want, become, online, gamer

Q. 21	Ans: (i) the total number of wrong predictions made by the model is the sum of false positive and false negative. $FP + FN = 25 + 15 = 40$ (ii) $Precision = TP / (TP + FP)$ $= 55 / (55 + 25) = 50 / 80$ $= 0.625$ $Recall = TP / (TP + FN)$ $= 55 / (55 + 15)$ $= 50 / 70$ $= 0.71$ $F1\ Score = 2 * Precision * Recall / (Precision + Recall)$ $= 2 * 0.625 * 0.71 / (0.625 + 0.71)$ $= 0.8875 / 1.335$ $= 0.67$ (1 marks for part (i) and ½ mark for each formula and ½ mark each for substitution of values in part(ii)) Please note: the mathematical calculations can be ignored	4
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